

Raja Chakraborty

Senior Product Engineer | Applied AI & Full-Stack Engineering

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PROFILE

Product engineer with 10+ years shipping production software across web, mobile, backend and business systems, with applied AI and recent native macOS work. Connects product requirements, interfaces, APIs, data, integrations, testing and operations from initial idea through verified release. Targets hands-on applied AI and product engineering roles building useful LLM-enabled applications.

Extensive product-management experience spanning product shaping, requirements, prioritization, cross-functional execution, QA and release coordination. Comfortable taking broad ownership in lean teams and translating business workflows into maintainable software.

APPLIED AI & PRODUCT CAPABILITIES

- **LLM-enabled applications:** LLM workflows, retrieval-augmented generation (RAG), prompt design, and agent/tool workflows; vector search concepts and AI-assisted automation.
- **Local AI and document processing:** on-device OCR, neural narration, forced alignment and synchronized text/audio experiences using SwiftUI, PDFKit, Vision, MLX and ONNX Runtime.
- **Production engineering:** authentication, REST APIs, background jobs, real-time workflows, integrations, data modeling, caching, release validation and production debugging.
- **End-to-end delivery:** requirements and edge cases, responsive UX, implementation, QA, deployment, ongoing operation and cross-functional coordination.

SELECTED AI-ENABLED PROJECTS

PDFVoice | Native macOS document-to-audio product

Private PDF audiobook reader with on-device OCR, neural narration, forced alignment and synchronized word highlighting. Built with Swift, SwiftUI, PDFKit, Vision, MLX and ONNX Runtime. Source: github.com/rjchakraborty/pdfvoice

GoalX | Local-first finance product with AI guidance

Flutter savings and goal tracker with manual accounts, recurring plans, versioned backup, optional owner-scoped Firestore sync and review-before-write AI guidance. Source: github.com/rjchakraborty/goalx

Robot/Shift | Learning platform with an AI tutor

Interactive robotics and ROS 2 field manual with portfolio projects, cloud progress and a security-bounded AI tutor. Next.js, React, TypeScript and Firebase. Source: github.com/rjchakraborty/robot-shift

TECHNICAL SKILLS

Web: React, Next.js, TypeScript, JavaScript, Tailwind CSS, Bootstrap

Backend and data: Node.js, Python, REST APIs, PostgreSQL, MongoDB, Firebase, Redis, Kafka

Mobile: Flutter, Dart, native Android, Java, Kotlin, store publishing

Native and AI: SwiftUI, PDFKit, OCR, MLX, ONNX Runtime, LLM/RAG and agent/tool workflows

Delivery: Git, CI/CD, testing, release validation, production support; AWS/GCP and Docker concepts

PROFESSIONAL EXPERIENCE

Full Stack Developer (Mobile & Web) | Ancile Technologies

November 2017 - Present

- Built and maintained mobile apps, responsive web apps, backend services, APIs, data models, integrations and production workflows.
- Owned delivery from requirements and UX through implementation, testing, debugging, deployment, release validation and ongoing operations.
- Delivered messaging, payment and location-service workflows using REST APIs, SDKs, Firebase, Maps and payment integrations.

Flutter and Native Android Developer | Guni - Engagement & Growth for SMBs

July 2016 - October 2017

- Built mobile products for SMB engagement and growth; integrated APIs, Firebase, Google Maps, push notifications, payments, messaging and location services.

Android Developer | RJS Tech Solution

May 2015 - June 2016

- Developed native Android applications and REST/JSON integrations; built and debugged services, content providers, broadcast receivers and location-aware workflows.

SELECTED SHIPPED PRODUCT EXPERIENCE

Classify: school planning, timetables, homework, tasks and progress statistics.

Shift Alerts: healthcare workforce scheduling, geolocation clock-in/out, messaging, voice calls, invoices and expenses.

Sav Money: savings goals, rewards, round-ups, payments and card-management workflows.

Tea ERP: payroll, purchasing, production, inventory, sales, maintenance and accounting.

EDUCATION & LANGUAGES

Post Graduate Diploma, Remote Sensing & GIS | Jadavpur University CAD Centre | 2015 - 2016

Bachelor of Arts, Geography | A.B.N. Seal College | 2012 - 2015

English: Fluent | Bengali: Native | Hindi: Native